

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – XI

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Implementing Advanced Visualization Techniques and Customizations

a) Create slope charts and lollipop charts

Step 1: Create Worksheet

- 1. Open Tableau Desktop and connect to Superstore.csv.**
- 2. Create a new worksheet.**
- 3. Rename it Slope Chart.**
- 4. Drag Order Date to the Columns shelf.**
- 5. Drag Sales to the Rows shelf.**

Step 2: Filter Years

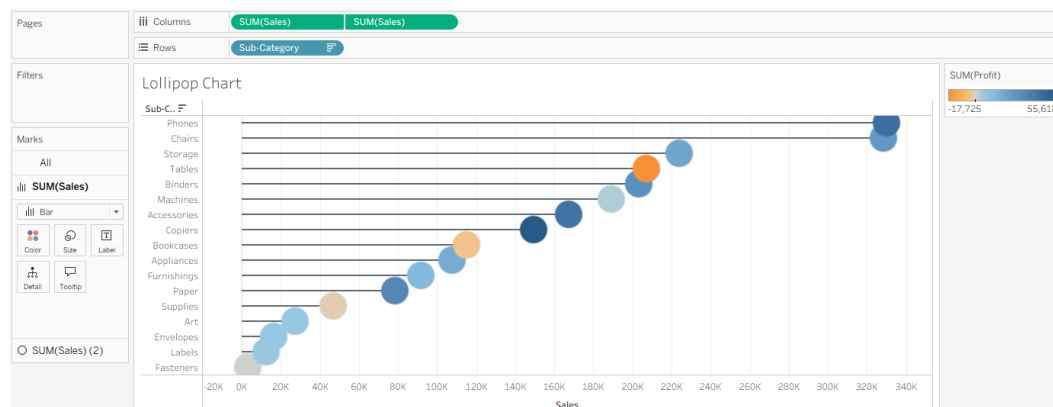
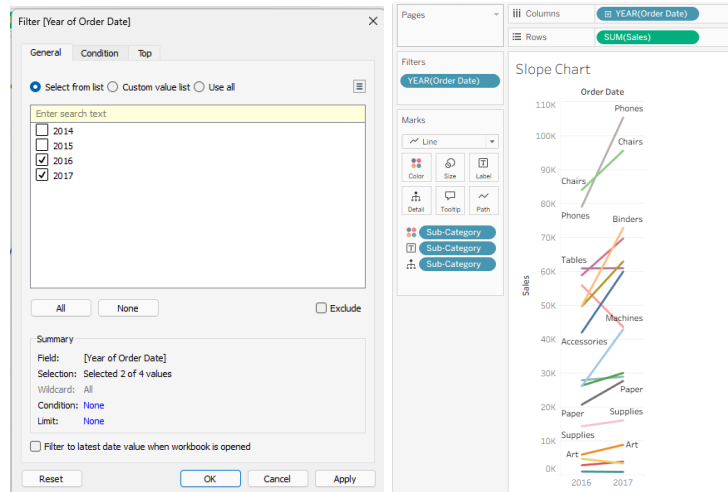
- 1. Drag Order Date to the Filters shelf.**
- 2. Select Years.**
- 3. Choose 2014 and 2017.**
- 4. Click OK.**
- 5. Right-click YEAR(Order Date) on Columns.**
- 6. Select Discrete.**

Step 3: Create Slope Lines

- 1. Change the Marks type to Line.**
- 2. Drag Sub-Category to Detail.**
- 3. Drag Sub-Category to Color.**

Step 4: Add Labels

- 1. Click Label on the Marks card.**
- 2. Check Show Mark Labels.**
- 3. Select Line Ends.**
- 4. Check Label Start of Line and Label End of Line.**



b) Design waterfall charts

Step 1: Create Worksheet

1. Create a new worksheet.
2. Rename it Lollipop Chart.
3. Drag Sub-Category to Rows.
4. Drag Sales to Columns.
5. Drag Sales to Columns again.

Step 2: Create Dual Axis

1. Right-click the second SUM(Sales).
2. Select Dual Axis.
3. Right-click the top axis.
4. Select Synchronize Axis.

Step 3: Configure Stem

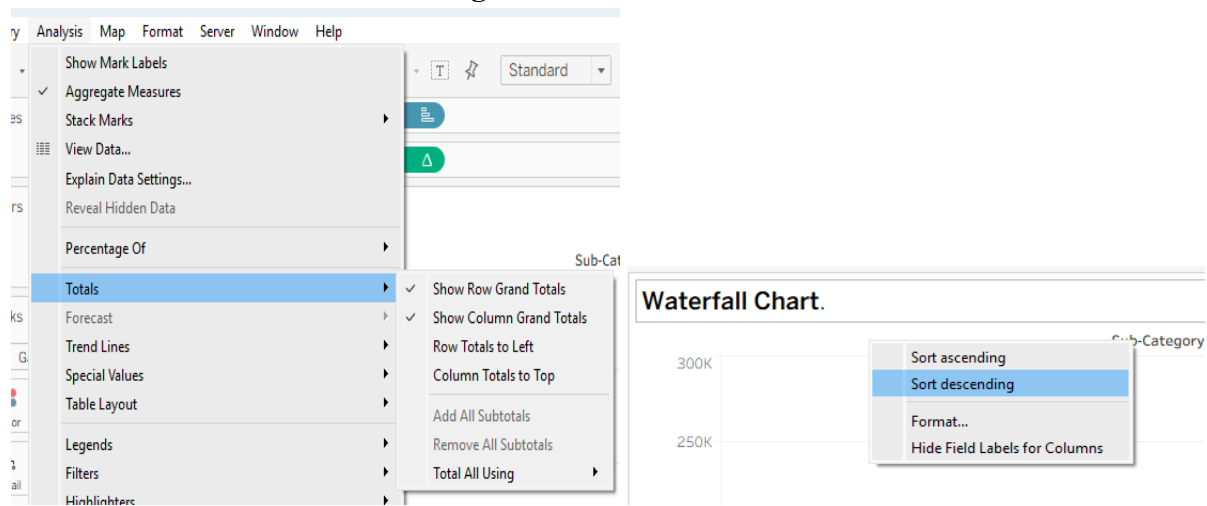
1. Select the first SUM(Sales) Marks card.
2. Change Mark Type to Bar.
3. Reduce the Size.
4. Change the bar color to Gray.

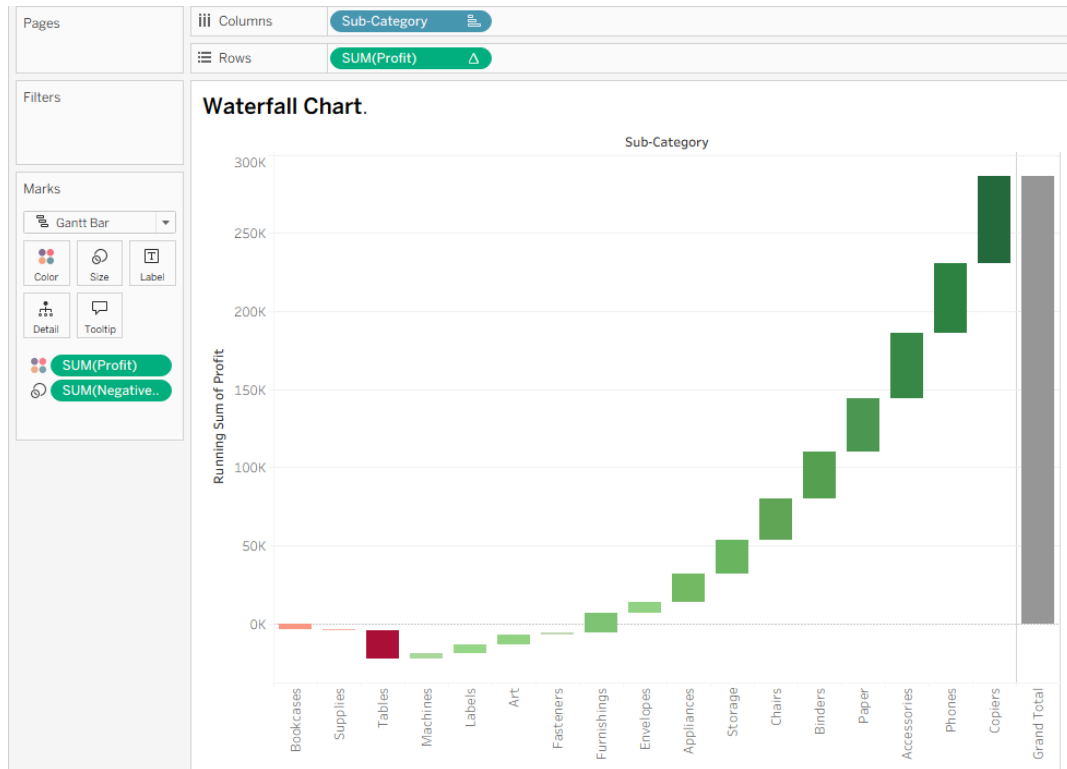
Step 4: Configure Head

1. Select the second SUM(Sales) Marks card.
2. Change Mark Type to Circle.
3. Increase the Size.
4. Drag Profit to Color.

Step 5: Format

1. Hide the top axis.
2. Sort the chart in descending order of Sales.





c) Build sparklines for trend analysis

Step 1: Create Worksheet

1. Create a worksheet named Sparklines View.
2. Drag Category to Rows.
3. Drag Sub-Category beside Category.

Step 2: Add Trend

1. Drag Order Date to Columns.
2. Select Continuous Month.
3. Drag Sales to Rows.

Step 3: Independent Axis

1. Right-click the Y-axis.
2. Select Edit Axis.
3. Choose Independent Axis Range for Each Row.

Step 4: Remove Clutter

1. Hide the X and Y headers.
2. Go to Format → Lines.
3. Set Grid Lines, Zero Lines and Trend Lines to None.
4. Remove Column Dividers.

Step 5: Highlight Min and Max

1. Change the line color.
2. Click Label.
3. Select Min/Max.
4. Enable minimum and maximum labels.

The screenshot displays the Tableau interface with three dialog boxes open for formatting a visualization.

Format Lines Dialog: This dialog is used to format various lines in the visualization. It includes tabs for 'Sheet', 'Rows', and 'Columns'. The 'Lines' section contains settings for 'Grid Lines', 'Zero Lines', 'Trend Lines', 'Ref Lines', 'Drop Lines', 'Axis Rulers', and 'Axis Ticks'. The 'Format Borders' dialog is also open, showing settings for 'Default', 'Total', 'Grand Total', 'Row Divider', and 'Column Divider'.

Format Borders Dialog: This dialog is used to format the borders of the visualization. It includes tabs for 'Sheet', 'Rows', and 'Columns'. The 'Default' section contains settings for 'Cell', 'Pane', and 'Header'. The 'Total' section contains settings for 'Pane' and 'Header'. The 'Grand Total' section contains settings for 'Pane' and 'Header'. The 'Row Divider' section contains settings for 'Pane', 'Header', and 'Level'. The 'Column Divider' section contains settings for 'Pane', 'Header', and 'Level'.

Marks to Label Settings: This section is located on the right side of the interface. It includes a checkbox for 'Show mark labels'. Below this, there are settings for 'Label Appearance' (Text, Font, Alignment) and 'Marks to Label' (All, Selected, Line Ends, Most Recent). The 'Scope' section includes settings for 'Table' and 'Cell'. The 'Field' section includes a dropdown menu for 'SUM(Sales)'. The 'Options' section includes checkboxes for 'Allow labels to overlap other marks', 'Label minimum value', and 'Label maximum value'.



d) Implement sheet swapping for dynamic dashboards

Step 1: Create Parameter

1. Right-click the Data Pane.
2. Select Create Parameter.
3. Name it Select_View_Parameter.
4. Data Type = String.
5. Allowable Values = List.
6. Add:
 - Category Sales (Bar)
 - Regional Profit (Map)
7. Click OK.
8. Show the parameter.

Step 2: Create Calculated Field

Create Show_Bar

[Select_View_Parameter]="Category Sales (Bar)"

Create Show_Map

[Select_View_Parameter]="Regional Profit (Map)"

Step 3: Create View 1

1. Create worksheet View 1 – Category Sales.
2. Drag Category to Rows.
3. Drag Sales to Columns.
4. Drag Category to Color.

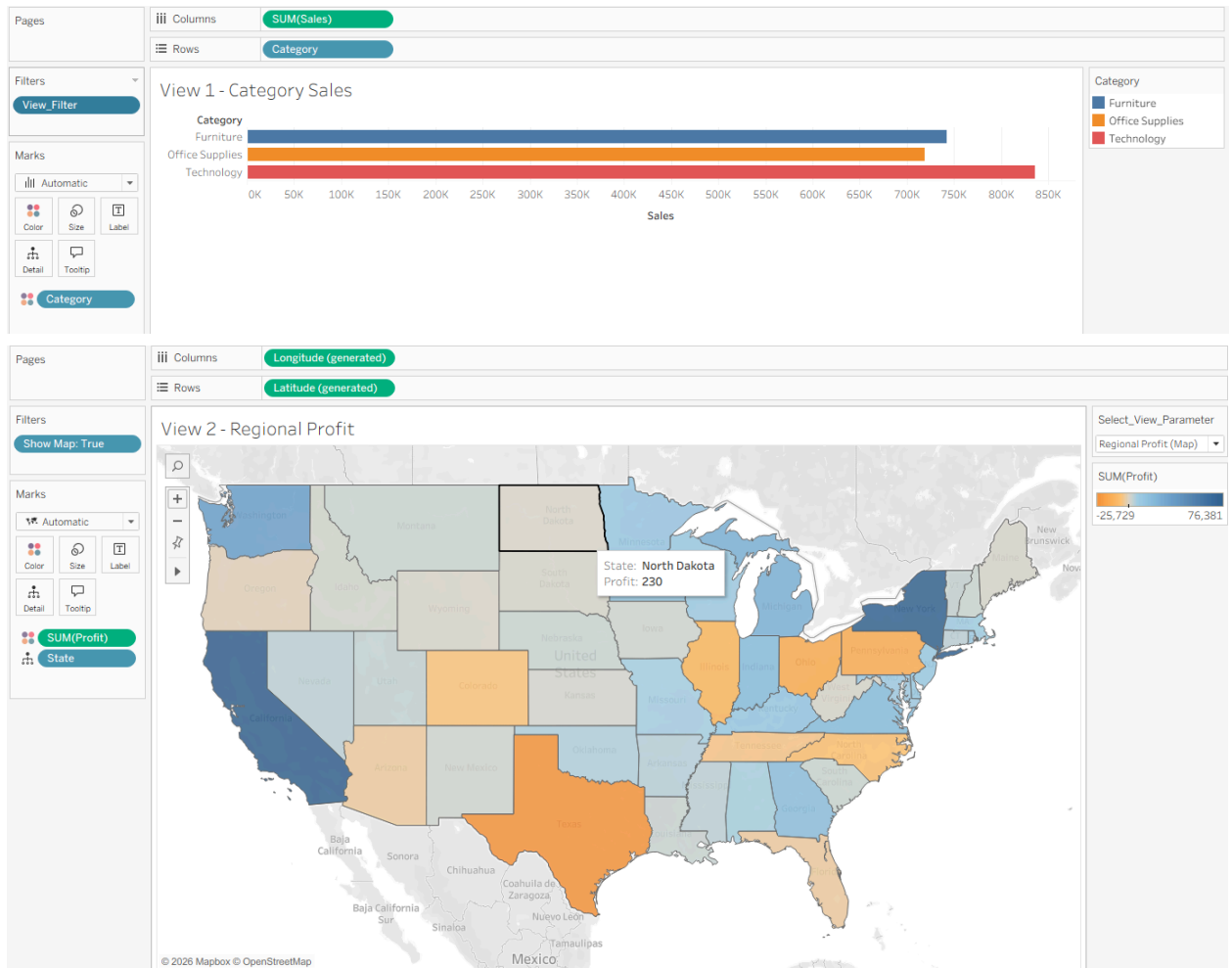
5. Drag Show_Bar to Filters.
6. Select True.

Step 4: Create View 2

1. Create worksheet View 2 – Regional Profit.
2. Double-click State.
3. Drag Profit to Color.
4. Drag Show_Map to Filters.
5. Select True.

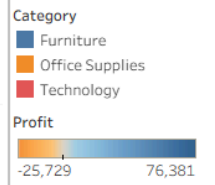
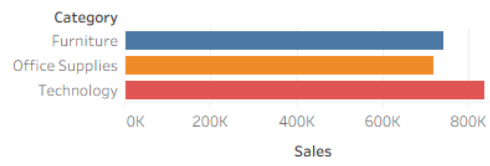
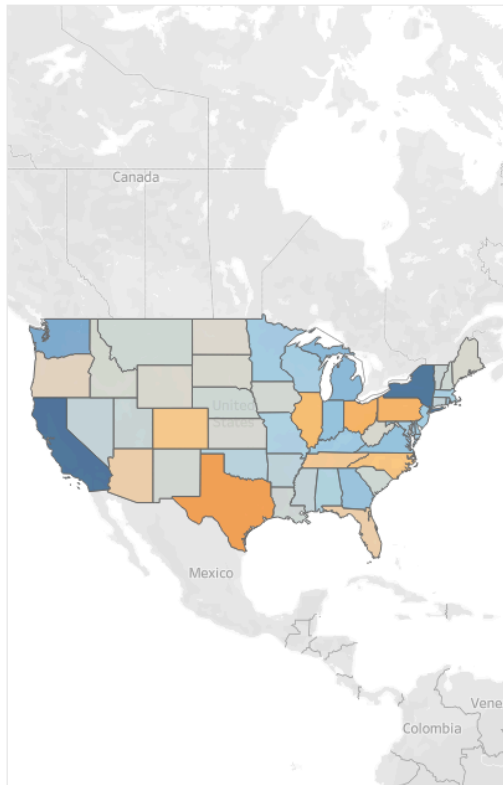
Step 5: Create Dashboard

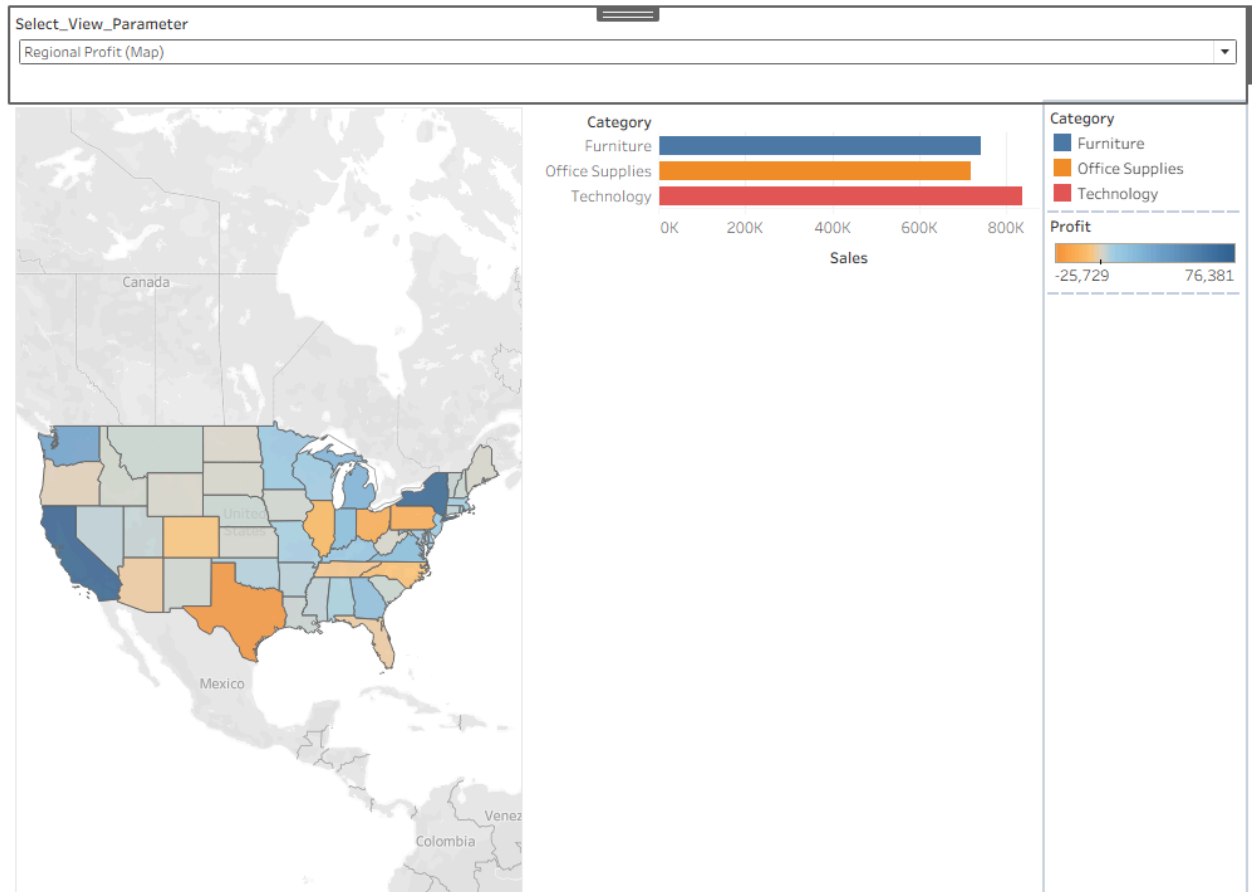
1. Create a dashboard.
2. Drag a Vertical Container.
3. Add both worksheets.
4. Hide worksheet titles.
5. Show Select_View_Parameter.
6. Test both parameter values.



Select_View_Parameter

Category Sales (Bar)





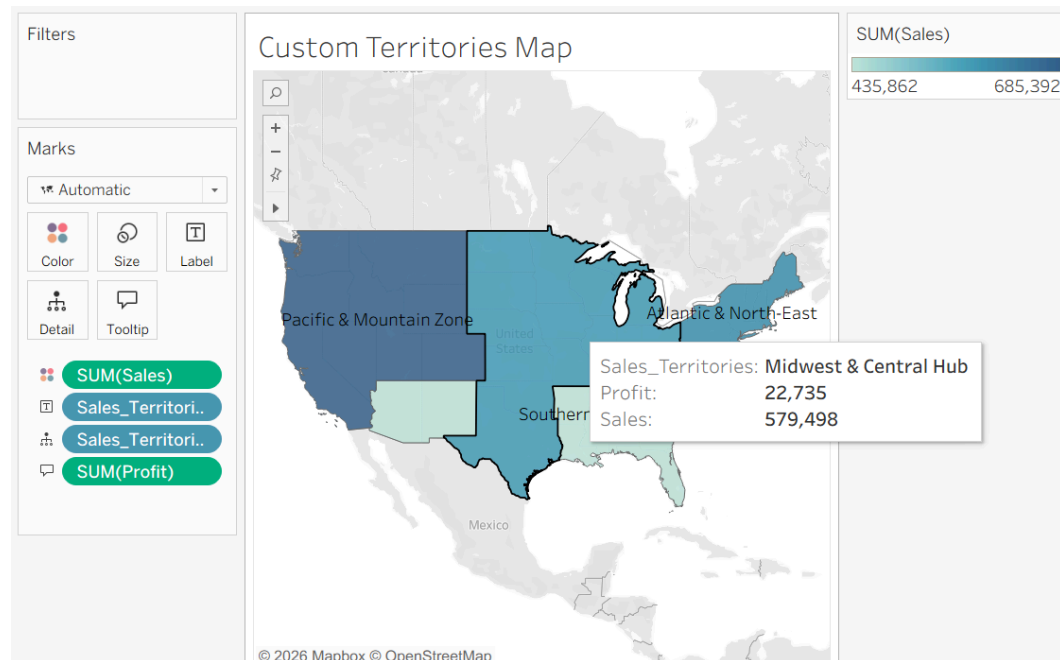
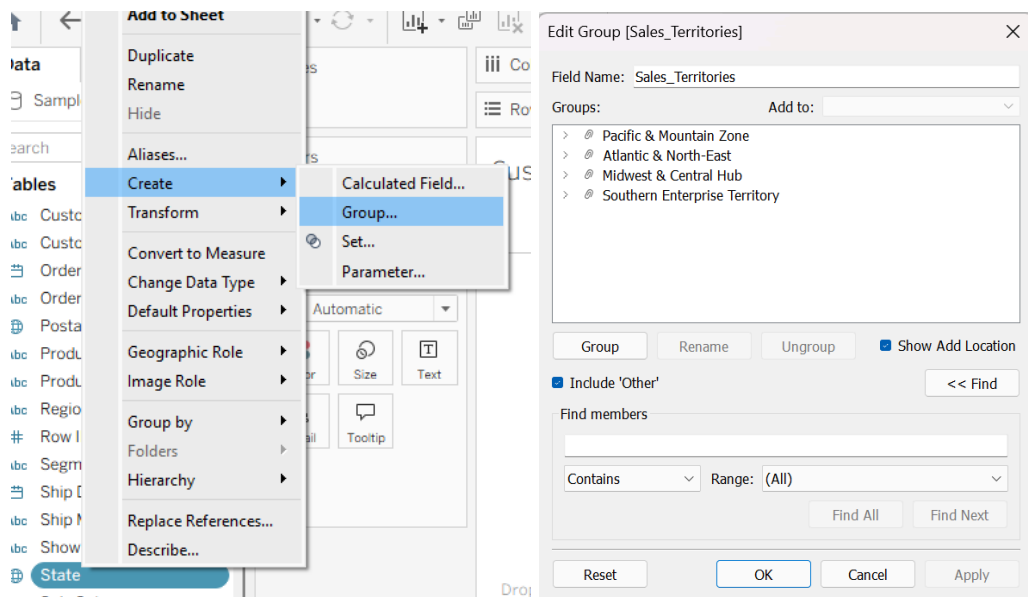
e) Create custom geographic territories

Step 1: Create Group

1. Create worksheet Custom Territories Map.
2. Right-click State.
3. Select Create → Group.
4. Group the required states.
5. Rename the groups:
 - Pacific & Mountain Zone
 - Midwest & Central Hub
 - Atlantic & North-East
 - Southern Enterprise Territory
6. Rename the group field Sales_Territories.

Step 2: Build Map

1. Double-click Sales_Territories.
2. Change Mark Type to Map.
3. Drag Sales to Color.
4. Drag Profit to Tooltip.
5. Drag Sales_Territories to Label.



f) Use background images in dashboards

Step 1: Create Coordinate Fields

Create Store_X_Coordinate.

(Create the CASE formula.)

Create Store_Y_Coordinate.

(Create the CASE formula.)

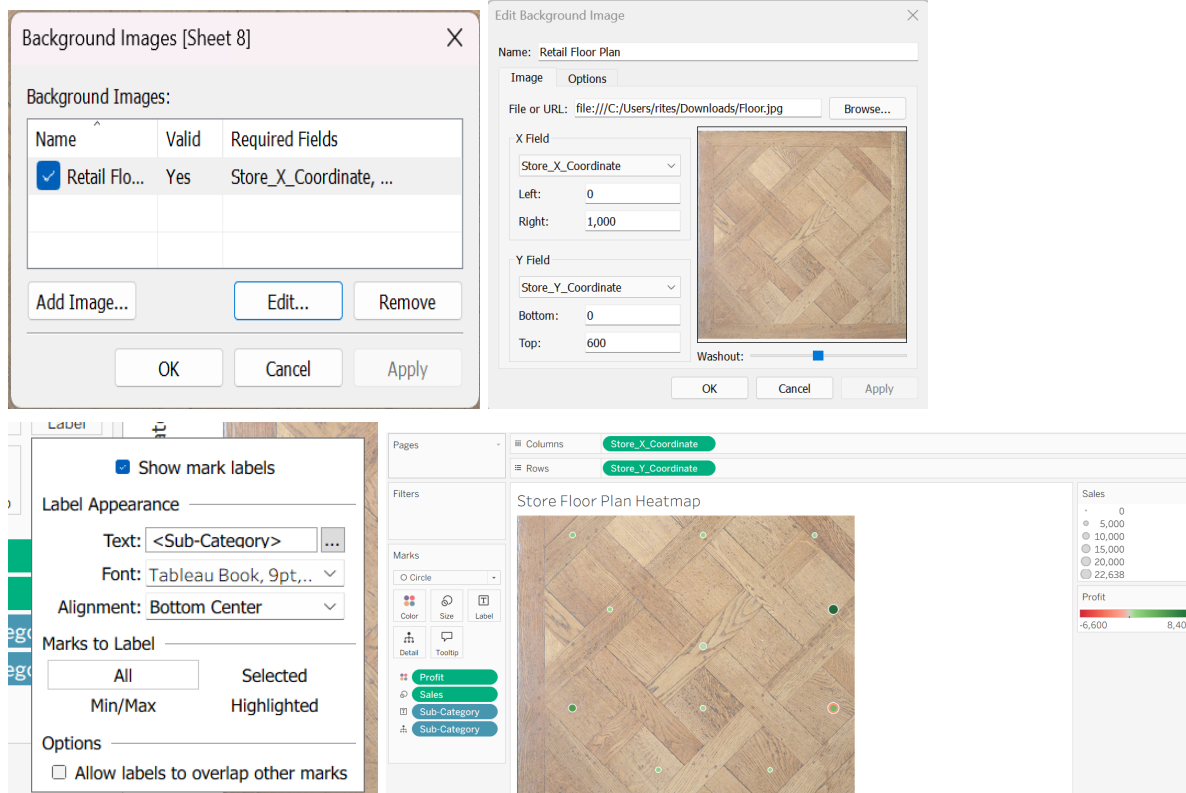
Step 2: Add Background Image

1. Go to Map → Background Images.
2. Select the data source.

3. Click Add Image.
4. Select the floor plan image.
5. Configure:
 - X Field = Store_X_Coordinate
 - Left = 0
 - Right = 1000
 - Y Field = Store_Y_Coordinate
 - Bottom = 0
 - Top = 600
6. Enable Lock Aspect Ratio.
7. Enable Always Show Entire Image.

Step 3: Create Visualization

1. Drag Store_X_Coordinate to Columns.
2. Drag Store_Y_Coordinate to Rows.
3. Drag Sub-Category to Detail.
4. Change Mark Type to Circle.
5. Drag Sales to Size.
6. Drag Profit to Color.
7. Drag Sub-Category to Label.
8. Hide both axes.
9. Remove grid lines.



g) Apply animation and transparency techniques

Task 1: Animated Motion Chart

Step 1: Create Scatter Plot

1. **Create worksheet Animated Motion Chart.**
2. **Drag Sales to Columns.**
3. **Drag Profit to Rows.**
4. **Drag Sub-Category to Detail and Color.**
5. **Drag Category to Shape (optional).**
6. **Change Mark Type to Circle.**

Step 2: Pages Shelf

1. **Drag Order Date to Pages.**
2. **Select Year (Discrete).**
3. **Display the Pages Control.**

Step 3: Motion History

1. **Enable Show History.**
2. **Set:**
 - **Marks = All**
 - **Show = Both**
 - **Fade = 50%**
3. **Set playback speed.**
4. **Click Play.**

Step 4: Workbook Animation

1. **Go to Format → Animations.**
2. **Enable Workbook Animations.**
3. **Set Duration to 0.5–1 second.**
4. **Set Style to Simultaneous.**

Task 2: Transparent KPI Card

Step 1: KPI Worksheet

1. **Create worksheet Transparent KPI Card.**
2. **Drag Sales to Text.**
3. **Format:**
 - **Font Size = 28 pt**
 - **Bold**
 - **White Color**

Step 2: Transparency

1. **Go to Format → Shading.**
2. **Set Worksheet = None.**
3. **Set Pane = None.**

Task 3: Dashboard

Step 1: Create Dashboard

1. **Create a new dashboard.**
2. **Rename it Animated & Transparent Master View.**

Step 2: Dashboard Background

1. Go to Format → Dashboard.
2. Choose a dark background color.

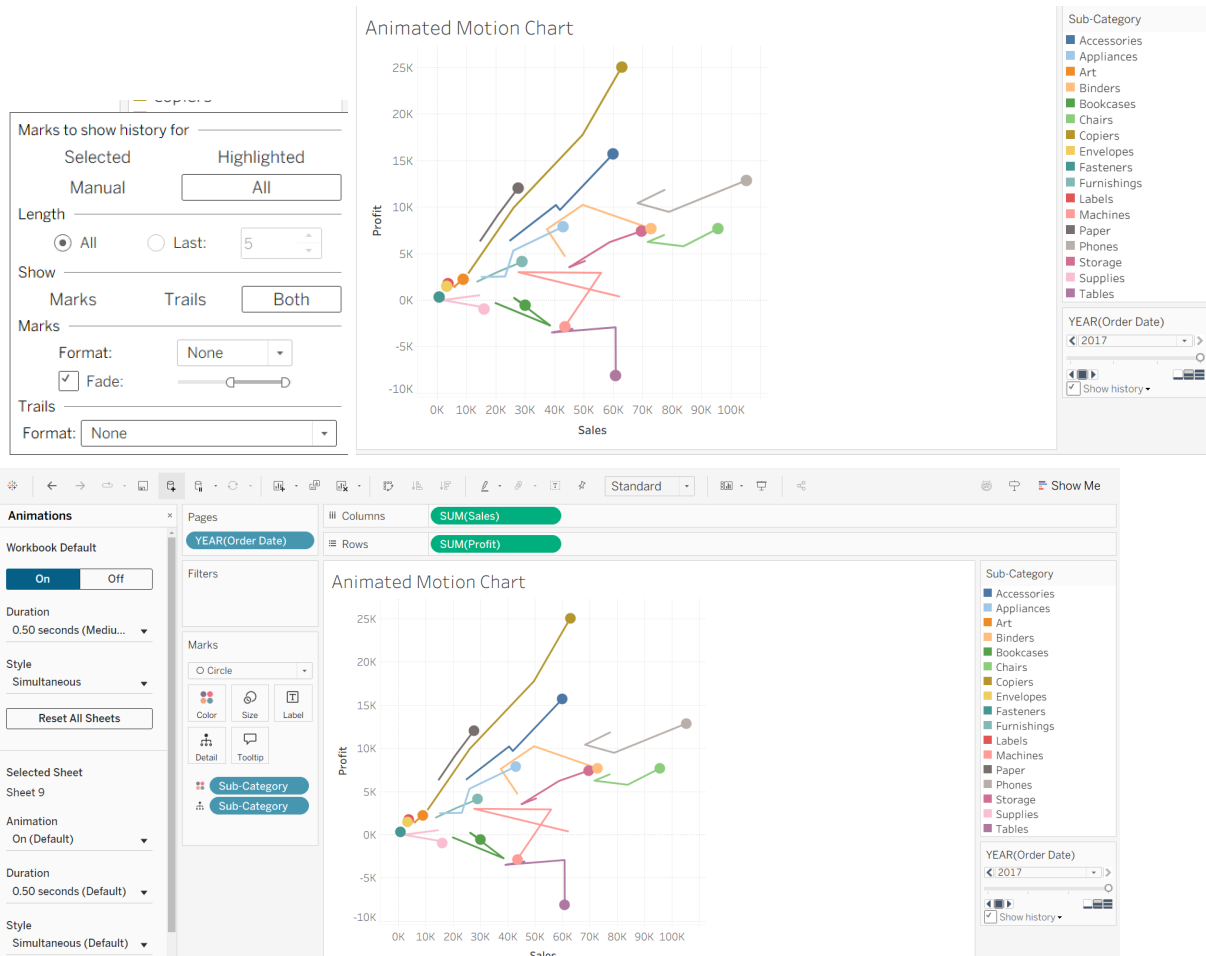
Step 3: Add Floating Sheets

1. Select Floating objects.
2. Add Store Floor Plan Heatmap or Custom Territories Map.
3. Add Transparent KPI Card.
4. Add Animated Motion Chart.
5. Arrange and resize the floating sheets.

Step 4: Test Dashboard

1. Click Play on the Pages control.
2. Verify:
 - Animated scatter plot.
 - Motion trails.
 - Transparent KPI card.
 - Background remains visible.

These steps are concise, documentation-ready, and suitable for inclusion in your practical record.



Sheet

ROWS

COLUMNS

Default

Worksheet:

None

Pane:

None

Header:

None

Pages

Columns

Rows

Filters

Marks

Automatic

Color

Size

Text

Detail

Tooltip

SUM(Sales)

Transparent KPI Card

2,297,201

Objects

- Horizontal Container
- Vertical Container
- Text
- Extension

Tiled

Floating

☒ Show dashboard title

Animated & Transparent Master View

Store Floor Plan Heatmap



- Sub-Category
- Accessories
 - Appliances
 - Art
 - Binders
 - Bookcases
 - Chairs
 - Copiers
 - Envelopes
 - Fasteners
 - Furnishings
 - Labels
 - Machines
 - Paper
 - Phones
 - Storage
 - Supplies
 - Tables

Year of Order Date

< 2017 >

Show history

Sales

0

5,000

10,000

15,000

20,000

22,638

Animated Motion Chart

